Krrish Kumar

Backend Engineer · Cloud & DevOps · ML Systems · Open to Internships & Startup Roles
Kolkata, India · +91 8697845183 · musashikk99@gmail.com · linkedin.com/in/krrishbuilds · github.com/Musashi-99

SUMMARY

18-year-old backend engineer with production-shipped **ML inference APIs, GPU simulation systems,** and cloud-deployed platforms. Proficient in **Python, FastAPI, AWS, Docker, LangChain.** Owns projects end-to-end — model training to REST endpoint to CI/CD deploy. Seeking **backend, DevOps, and ML internships** and startup roles where ownership matters.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++
Backend: FastAPI, Node.js, Express.js, REST API Design, JWT, Microservices
Cloud/DevOps: AWS (EC2, S3, IAM), GCP, Docker, Linux, CI/CD, GitHub Actions, Vercel, Infrastructure as Code
ML / AI: Scikit-learn, XGBoost, SHAP, Pandas, NumPy, LangChain, OpenAI API, Claude API, n8n
Databases: PostgreSQL, Redis, MongoDB
Frontend: React, Next.js, Tailwind CSS, Three.js, WebGL, GLSL, MediaPipe

PROJECTS

SME Credit Risk Engine | *Python · XGBoost · SHAP · FastAPI · PostgreSQL*

- Designed end-to-end ML credit scoring pipeline using **XGBoost** achieving **AUC-ROC of 0.89** on holdout set, with per-prediction **SHAP explainability** for interpretable, production-ready inference
- Engineered feature pipeline over financial and behavioral SME data; exposed predictions via **validated FastAPI REST endpoints** with structured JSON responses and error handling
- Architected **modular system** separating model, pipeline, and API layers — built for **zero-downtime hot-swapping** and production extensibility

IntelliMaintain | *Python · Scikit-learn · FastAPI · PostgreSQL*

- Built predictive maintenance system on time-series sensor telemetry, detecting anomalies with **under 5% false-positive rate** using statistical feature extraction and threshold alerting
- Developed **REST API handling 200+ req/min** with **<80ms average latency**, exposing real-time predictions for IoT monitoring dashboards and external alerting pipelines
- Enforced clean separation of **data ingestion, inference, and alerting layers** for maintainable, production-grade backend architecture

NeuroMorph Engine | *Three.js · WebGL · GLSL · MediaPipe · TypeScript*

- Engineered **GPU particle simulation** with custom GLSL shaders sustaining **5,000+ particles at 60fps** in-browser — zero backend, no install required
- Integrated **MediaPipe hand landmark tracking** to translate live gestures into real-time Three.js scene transformations via **client-side GPU compute**

Orbito | *Next.js · React · Tailwind CSS · TypeScript · Vercel*

- Built a **multi-category service marketplace** using Next.js with **SSR, dynamic routing,** and a reusable Tailwind component system across 10+ service pages
- Configured **CI/CD pipeline via Vercel** with auto-deploys on every push to main; **zero-downtime production deployments** since launch

EDUCATION

B.Tech in Cybersecurity — Expected May 2029
University of Engineering & Management (UEM), Kolkata

Senior Secondary — Science (PCM) — 2025
Kendriya Vidyalaya

ACHIEVEMENTS & CERTIFICATIONS

- **East India Blockchain Summit 2.0** — Finalist
- **IIT Kharagpur Data Science Hackathon 2026** — Participant
- **Deloitte Cyber Job Simulation** — Forage
- **HCL GUVI Certificate** — AI Impact Track
- **Buildathon** — Participant · **Convolve 4.0** — Participant